Day 1: Introduction to Networking and the OSI Model

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Course: Diploma in Cyber Security

Topic: Day 1: Introduction to Networking and the OSI Model

Introduction

Networking means connecting computers, phones, servers and other devices so they can exchange data. We use networking every day when browsing websites, sending WhatsApp messages, watching videos or using online services.

The OSI model is a seven-layer model used to understand how data moves from one device to another. Each layer has a different job, which makes networking easier to learn and troubleshoot.

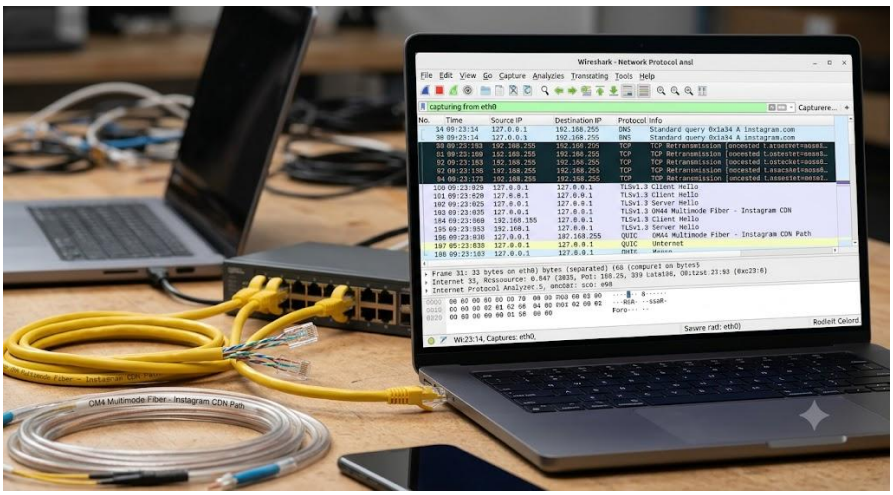
Task 1

PRACTICAL

What I have to do:
1. Install Wireshark from the official Wireshark website.
2. Open Wireshark and select the main network interface, such as Wi-Fi or Ethernet.
3. Start the capture.
4. Open a browser and visit instagram.com.
5. Wait for the page to load, then stop the capture.
6. Look at the Protocol column and write the protocol names that actually appear in the capture.

Software/tool: Wireshark and a web browser.

Result to record: The protocol names shown in your own packet list.



Important: Do not invent the protocol list. It depends on the computer, browser, network and website. Use your own capture.

Task 2

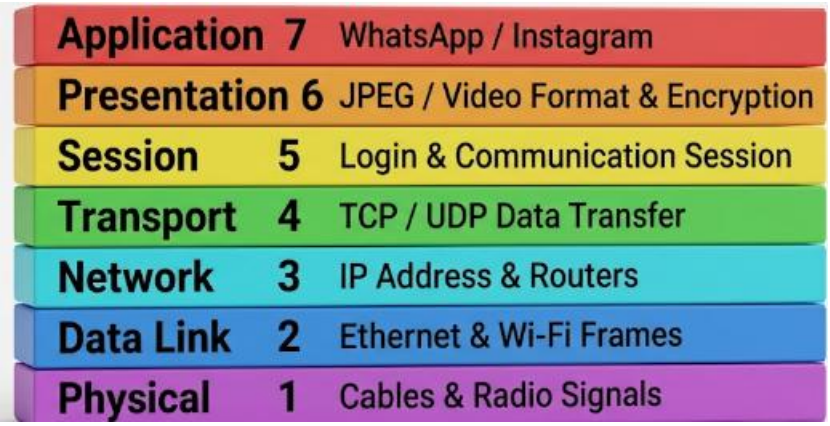
PRACTICAL

Pick three packets from your own Wireshark capture. Use the Protocol and Info columns. A suitable identification can look like this:

Example Protocol	**Main OSI Layer**	**Why**
HTTP/HTTPS	Layer 7 – Application	Carries web application data.
TCP	Layer 4 – Transport	Provides transport and connection management.
IP/IPv6	Layer 3 – Network	Provides logical addressing and routing.

These are examples only. Your actual three packets must be identified from your own capture.

Task 3



Draw a seven-layer OSI model from Layer 7 at the top to Layer 1 at the bottom. Use different colours for the layers. Add one simple daily-life example beside each layer.

Suggested content:
Layer 7 – Application – WhatsApp / Instagram
Layer 6 – Presentation – image or video formatting, encryption
Layer 5 – Session – maintaining a communication session
Layer 4 – Transport – TCP/UDP
Layer 3 – Network – IP address and routing
Layer 2 – Data Link – Ethernet / Wi-Fi frames
Layer 1 – Physical – cables, radio signals

If you make it on paper, use [PHOTO TO ADD HERE] and add a clear photo of your completed diagram.

Task 4

When I send a WhatsApp message, the message passes through several networking layers. At the Application layer, I type the message and WhatsApp prepares it for communication. At the Presentation layer, data can be formatted and encrypted so it can be safely handled.

At the Transport layer, the data is divided and managed for delivery. At the Network layer, IP addressing helps the data find the destination network. Then the lower layers put the data into frames and send it through Wi-Fi or another physical connection. On the receiver's side, the process works back up the layers and the message appears in WhatsApp.

Conclusion

The OSI model gives a simple way to understand network communication. Wireshark can show real network packets, while the seven layers help us understand what each part of the communication is doing.

Quick Revision

- OSI has 7 layers.
- Layer 7 is Application and Layer 1 is Physical.
- Wireshark can be used to observe real traffic.
- TCP mainly belongs to the Transport layer.
- IP mainly belongs to the Network layer.